Final Exam: Information Integration and Applications (CSE656)

AY: 24-25, Monsoon Semester

Time: 2.0hrs

Marks: 30

Note: Please state your reasonable assumptions clearly, if any, and justify why you have taken these assumptions. Your assumptions should not contradict anything stated in the problem statement/question.

There are four short-answer questions (based on case study ONE) and 40 MCQs (based on case study TWO). You can either attempt (Q1, Q4 and MCQs) or (Q1, Q2, Q3 and Q4). If you decide to attempt MCQs, you can use your laptop, and the Google form Link will be shared. You need to decide this within 5 mts after distributing the paper (without seeing the options on Google form). For any reason, if your laptop is not working, you need to attempt (Q1, Q2, Q3 and Q4).

Case Study ONE

Imagine you are starting a new start-up, called 'SastaBookStore.com'. The goal of your start-up is to build a platform that (1) provides a seamless shopping experience by presenting available books from all its bookstore partners in a consistent and coherent manner when a (potential) customer poses a query, (2) provides a customer feedback system where the registered customer may provide the feedback of a book. This feedback contains the feedback category from a drop-down list (like, book review, complaint, error reporting, etc), and book title, and the textual description of her feedback (limit to 200 characters). This feedback information gets persisted in a relational database, and (3) provides a preface of each book (ISBN, Title, Edition Number) and the reviews from a third party (BookPreface.com) to its potential customers and this information gets materialize in a database linked to a local file system of a SastaBookStore.com for each book, if it is not locally available.

Let's design a part of integrated information system as given below:

You are tasked with **integrating underlying bookstores schemas and BookPreface.com** from three online bookstore platforms and from 'BookPreface.com' into a unified set of schemas. Further, you have been asked to classify each book into only one subject area from a given subject areas list and materialize the title of the book and its category. For example, a book titled 'Fundamentals of Database Systems' is classified into the subject area 'Information Management'. Each book title is classified once. The unregistered customers can search the book by (and/or) title/author/subject-area/publisher/year of publication, and also view the book-preface and reviews before they decide to buy a book or not. However, when a customer wishes to buy a book, (s)he needs to register on the platform before executing a transaction. Each transaction is recorded in the database (customer info, bookstore info, ISBN, price, purchase date and time).

The 'SastaBookStore.com' also builds a **data warehouse** to track the sales by bookstore, by classification of books and by time, and does aggregate sales analysis. It is being used by the business decision makers.

Each underlying bookstore exposes the following schema to the 'SastaBookStore.com'. Note, each bookstore may have a different price for the same book. The primary key in each schema is underlined.

BookStore1(<u>ISBN</u>, Title, <u>Pub-Year</u>, Availability, Unit_Price)

BookStore2(<u>ISBN</u>, BTitle, <u>Year-of-Publication</u>, Inventory, Price)

BookStore3(<u>ISBN</u>, Book_Title, <u>Publication-Year</u>, InStock, Price)

BookPreface.com(<u>ISBN, Edition#,</u> User-Avg-rating, Preface, Reviews)

[Note BookPreface data is not a complete structured data, thus not stored in a relational table(s))

Here, the values of Availability/Inventory/InStock are {Yes, No}. Note the ISBN of a book does not change even if the edition number changes.

Q1: For the above problem statement,

a) Design the **global schema** of the 'SastaBookStore.com' where a (potential) customer can pose a query (1) to find out the availability and price of the book at each BookStore, (2) to view the preface and review of a book by ISBN/Title, (3) to view the different editions/year of publications of a book (ISBN/Title) available, (4) to view the user-avg-rating of a book by title and year of publication), and (5) to view the customers transactions. If the book is not available in a BookStore, the result of that BookStore should not be shown to the client. (2)

b) Write the schema mapping between the global schemas (obtained in part (a)) and local schemas using GAV. (2)

c) The SastaBookStore.com decides to show the lowest price of each book, that is, one record per ISBN unless more than one BookStore has the same lowest price. For this constraint, design the schema and its mapping to global schema (obtained in part (a)). (2)

d) If you decide to take a materialized approach for designing the above system for book search/preface/reviews, what would be your architecture and what challenges you might face in designing such a system? Will you suggest such architecture, as a database expert, and why? (2)

Attempt either (Q2 and Q3) or MCQs on Google Form:-

Q2: For the customer feedback database described above, assume the 'SastaBookStore.com' has sold more than a million books as of today and they have received more than 0.5 million feedbacks. It is possible that some people may provide the feedback (irrespective of its category) even if they have not bought the books through the 'SastaBookStore.com'.

a) Design the customer feedback database schema. (3)

b) Write your design methodology to correlate each feedback with the sold book ISBN. Remember, the title provided by the user in the feedback form may be incomplete/erroneous/does not match at all with the sold book titles. (4)

c) Describe your methodology using an example and demonstrate that your design is scalable for correlating millions of structured records with customer feedbacks. (4)

Q3: In the above problem statement, the SastaBookStore.com generates revenues in two different ways, namely (1) 3% from each transaction, and (2) 5% from the total revenue, if it exceeds more than Rs10L per year per BookStore. The SastaBookStore.com stores and manages all sales transactions. To manage the book sales by BookStore (BookStore-Name, City, State) and Time (Date, Month, Year), the business decides to build a data warehouse (DWH) which maintains a materialized view storing the total revenue and total number of books sold by BookStore and by Date.

Note: the client will interact with the SastaBookStore.com for any transaction including returning a book.

a) What would be the facts and dimension tables in this scenario? Write the schema for each. (1+1)

b) Write an OLAP query for computing the total sales by BookStore, Date? (1)

c) Discuss your method to maintain the above materialized view incrementally whenever a customer returns a book. (1)

Q4: Let us assume each BookStore1 and BookStore2 businesses have their own registered customers. For business reasons, The BookStore1 acquired the business of BookStore2, and therefore the BookStore2 registered customers profiles will be merged into the profiles of BookStore1 registered customers. It may happen that two different customers from BookStore1 and BookStore2 may have the same username. To avoid this clash, the BookStore1 would like to know the common usernames in both BookStores. Discuss your query method using Bloom Join to find out the common usernames. [Note: Only Bloom join definition will attract Zero marks. You need to discuss the method w.r.t. this given scenario]. (2)

Case Study TWO

A 'Charming Bank' (CB) has several branches in a city which can be managed by a person. It offers retail banking products/services to customers. The 'Multiply Financial Services' (MFS) offers different investment and insurance related services and products to their high net worth individuals, and has one office per city. The 'Zero Brokerage Financial Ltd' (ZERO) offers an online platform where the clients can trade the equities with zero brokerage. The ZERO partners with the 'Charming Bank' for their customers' banking needs, that is, linking the customers trading account with the bank account (for these customer, the CustomerID is prefix with '00' in the following Customer_Accounts table). Thus, the CB has two categorizes of customers, namely, their own customers and the customers of ZERO. A customer can have accounts in all CB, MFS and ZERO.

The CB management board decides to expand the business by offering services and products related to the financial sectors, and therefore decides to acquire the MFS so that the CB can reach out to broader market to acquire new clients and also do cross-offering services and products to the all customers. To achieve this objective, the CB builds an information integration system, for the following applications, which integrates data from newly acquired MFS into the CB system to provide a unified view of customer accounts, transactions, and branch information. The applications are:

- **Unified Customer View**: Consolidate customer profiles across banks.
- **Transaction History**: Merge transaction records from different banks.
- **Branch Information**: Integrate and standardize branch details from acquired banks.
- **Customer Financial Overview**: Provide a comprehensive financial summary for each customer.
- **Fraud Detection**: Aggregate transaction data for detecting fraudulent activities.

Taking a simpler view of the above problem, the schemas of the local databases and the mediated schemas are given below.

The CB database schemas are (The primary key is each schema is underlined):

Customer_Accounts(CustomerID, FullName, AccountNumber, BranchCity, Balance)

Transactions(TransactionID, AccountNumber, BranchID, CustomerID, Product_Name, Transaction_Date, Amount, Trans_Type)

Branches(BranchID, BranchCity, Managed_By)

The MFS database schemas are (The primary key is each schema is underlined):

Customer_Profile (ClientNumber, Name, AccountCode, OfficeCity, Invst_Amount)

Products_Transactions (TransNum, AccountCode, OfficeCode, ClientNumber, ProductName, Date, Amount, BuyORSell)

Offices (OfficeCode, OfficeCity, ManagerName)

The ZERO database schema is (The primary key is the schema is underlined):

Customer_Profile (CB_BankAccountID, GuestNumber, Guest_Name, Guest_City, Portfolio_Value)

Mediated Schemas

Unified_Customers(CustomerID, FullName, AccountNumber, BranchCity, Balance)

Unified_Transactions(TransactionID, AccountNumber, BranchID, CustomerID, ProductName, TransactionDate, Amount, TransactionType)

Unified_Branches(BranchID, BranchCity, Branch_Managed_By)

Materialized Views Schemas

Customer_Financial_Overview_View (FullName, AccountNumber, Total_Net_Worth)

Branch_Transaction_Summary (BranchName, TotalTransactions, TotalAmount)

The data instances of the following relational tables are:

CB: Customer_Accounts

CustomerID	FullName	AccountNumber	Balance	BranchCity
1	Ramesh Jaju	235	100000	New Delhi
2	Deba Jyoti	324	340000	New Delhi
003	Venkata R	346	560000	Lucknow
4	Ratan Bihari	321	5000	Patna

MFS: Customer_Profile

ClientNumber	Name	AccountCode	OfficeCity	Invst_Amount
C11	Rakesh Jaoo	S2353	New Delhi	5000000
C12	Deva Jyothi	S2373	New Delhi	9400000
C13	Venkat Ramaswamy	S4321	Kochi	9600000
C14	Rathan Vihaari	S5462	Bengaluru	1000000

Q5: Need to attempt this question on Google form link that will be shared during the exam. Select the correct answer(s). Based on above case study.

220112. Which string matching algorithm would be most suitable to match the full names "Ramesh Jaju" and "Rakesh Jaoo"?

356B81. When matching customer names like "Deba Jyoti" vs. "Deva Jyothi," which of the following technique(s) is/are most appropriate for detecting subtle name variations?

A78B41. In the context of address matching, if two addresses differ only in the order of street and neighbourhood names (e.g., "12, Lakshmi Vilas, Arya Nagar Colony" vs."12, Arya Nagar Colony, Lakshmi Vilas"), which string similarity algorithm(s) would be the best choice for matching them?

Q789U1. Which of the following would be the most effective rule-based approach(es) for resolving discrepancies in customer addresses such as "Shaheed Nagar, Ashoka Marg, Agra" vs. "Shaheed Nagar, Ashoka Road, Agra"?

WER671. What is/are the main challenge(s) in matching customer names like "Ratan Bihari" vs. "Rathan Vihaari"?

2347WE. If you are using supervised learning to match customer profiles based on names
and addresses, which of the following would be the most useful feature(s) for the classifier to focus on?

TUY342. What is/are the main advantage(s) of using a learning-based algorithm over rule-based matching for customer entity resolution?

562UV2. Which of the following would be a disadvantage(s) of using exact string matching in customer entity resolution?

U345T2. Given documents, what artifacts we can extract from these documents which is/are required for entity matching and/or correlating (structured) data with documents and/or even for documents similarity.

789BU1. In a dictionary based approach for entity extraction, which method(s) we leverage for person-name extraction?

23489B. Which of the following similarity measure(s) can we use for data string matching?

184BN3. Which of the following techniques (or in combination of) can be used for schema matching?

4BA627. Which SQL statement creates a table to store unified customer data?

781C25. In the given above schema of **Branch_Transaction_Summary**, how many Dimension Tables (DT) and Fact Tables (FT) would be required to create a materialized view that aggregates transaction amounts and total number of transactions by branch name?

893B67. What would be the super key of **Unified_Branches** relation?

782B79. Which of the following are correct mappings for integrating CB.Branches and MFS.Offices schemas?

56B78G. When the above mediated schemas are designed, what aspects should be considered?

31B789. What is the main purpose of mapping schemas of **Customer_Accounts** and **Customer_Profiles** in the integration process?

789T12. Which attributes should be used as Blocking and Matching for entity matching in CB.Customer_Accounts and MFS.Customer_Profile tables (assume the nomenclature of attributes in both schemas is same)?

189B32. What is the primary function of the `Unified_Branches` table in the mediated

schema?

7BHU29. How can missing data affect the entity matching process?

256BN0. Which techniques are useful for schema matching when integrating data from

heterogeneous sources?

6NUTB1. In evaluating an entity resolution process, which of the following are true regarding

precision and recall?

89GH27. Which strategy would be better to match the 'FullName' attribute values of **Customer_Accounts** with the 'Name' attribute values of **Customer_Profile** data strings?

289BG8. To design the mediated schemas of unified customers, transactions and branches, which design methodology would be more applicable?

2BHU81. Which of the following algorithms may be used for 'FullName' and 'Name' string

similarity measurement?

89G674. If we need to find those customers who are having a/cs in both CB and MFS, what

method(s) would you use?

89UT31. Let's say the Customer_Accounts table has 100 million records and

Customer_Profile table has 10 million records. What challenges you may encounter in for Entity Matching task?

278BH2. What challenges might arise in schema mapping when integrating customer data?

41G678. What is the main advantage of **Customer_Financial_Overview_View** and **Branch_Transaction_Summary** materialized views in the integrated system?

11987T. In the above schema integration, which approach is best for handling synonyms in
field names (e.g., `FullName` vs. `Name`, 'BranchID' vs 'OfficeCode')?

89BH26. What would be the Jaro Score for matching 'Deba Jyoti' and 'Deva Jyothi' strings in the above CB.Customer_Accounts and MFS.Customer_Profile relations? The formula to compute the Jaro score is $jaro(x,y) = 1 / 3[c/|x| + c/|y| + (c - t/2)/c]$. Please do not consider space as a character in the both strings.

613B78. In the above mediated schemas, how is `AccountNumber` used to integrate
transactions?

56BG17. What is a primary benefit of using '**Customer_Financial_Overview**' materialized views for reporting?**

6HU219. What will be the most suitable architecture for the above information integration
system?

91245G. We know that each Branch belongs to a District, a District belongs to a state. This information can be captured in a schema **Branch-Info**(BranchName,District, State). Which SQL statement will analyse the total business done by all branches in a district and in a State?

978V45. In the above data instances, which method is commonly used for Name string
matching to align fields with similar names?

789B62. In the above CB database schemas, how can schema mapping help in dealing with inconsistencies in data formats across integrated sources?

4BH23I. Which methods can be used to handle variations in names during entity matching? **6NH241**. How can domain-specific knowledge assist in schema matching?